



Subject: Engineering Mathematics-I

Subject Code: MAL101

Branch: IT

Max.Marks.-50

Time- 3 Hours

Note: All questions are compulsory. (Rough sketch compulsory in Q-4).

Q-1	Show that the matrix $A = \begin{bmatrix} 3 & 1 & -1 \\ -2 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$ is diagonalizable. Hence, find the modal matrix P such that $P^{-1}AP$ is a diagonal matrix. And also obtain the matrix $B = A^2 + 5A + 3I$.	5 Marks
Q-2	(a) Reduce the matrix $A = \begin{bmatrix} 2 & 1 & 3 & 4 \\ 0 & 3 & 4 & 1 \\ 2 & 3 & 7 & 5 \\ 2 & 5 & 11 & 6 \end{bmatrix}$ to normal form and hence find its rank. (b) Find the value of λ such that the system of homogeneous equations $2x + y + 2z = 0$, $x + y + 3z = 0$, $4x + 3y + \lambda z = 0$ has (i) Trivial solution (ii) non-trivial solution. Hence, find non-trivial solution.	3 Marks 2 Marks
Q-3	(a) Verify Rolle's theorem for the function $f(x) = (x-a)^m(x-b)^n$, where m and n are positive integers on $[a, b]$. (b) find the coordinates of the centre of curvature at the point $(1, 1)$ on the curve $x^3 + y^3 = 2$.	2 Marks 3 Marks
Q-4	Trace the curve $y^2(a+x) = x^2(a-x)$, $a > 0$.	5 Marks
Q-5	(a) Show that the function $f(x, y) = \frac{xy}{x^2+2y^2}$, when $(x, y) \neq (0, 0)$ and equal to zero when $(x, y) = (0, 0)$ is not continuous at $(0, 0)$, but its partial derivatives f_x and f_y exist at $(0, 0)$. (b) find the equation of the tangent plane and normal line to the surface $x^2 + 2y^2 + 3z^2 = 12$ at $(1, 2, -1)$.	2 Marks 3 Marks
Q-6	If $\frac{x^2}{a^2+u} + \frac{y^2}{b^2+u} + \frac{z^2}{c^2+u} = 1$, then prove that $(u_x)^2 + (u_y)^2 + (u_z)^2 = 2(xu_x + yu_y + zu_z)$.	5 Marks
Q-7	(a) If $u = \frac{x^3y^3z^3}{x^3+y^3+z^3} + \log\left(\frac{xy+yz+zx}{x^2+y^2+z^2}\right)$, then prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = \frac{6x^3y^3z^3}{x^3+y^3+z^3}$. (b) If $x = e^{\theta+i\varphi}$ and $y = e^{\theta-i\varphi}$, then show that $\frac{\partial^2 u}{\partial \theta^2} + \frac{\partial^2 u}{\partial \varphi^2} = 4xy \frac{\partial^2 u}{\partial x \partial y}$, where u is a Composite function of θ and φ .	2 Marks 3 Marks
Q-8	Find volume of greatest rectangular parallelopiped can be inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.	5 Marks
Q-9	(a) Find the volume of the solid generated by revolving an arch of the cycloid $x = a(t - \sin t)$, $y = a(1 - \cos t)$ and x-axis about the x-axis. (b) find the surface area of the solid generated by the revolution of the asteroid $x = a \cos^3 t$, $y = a \sin^3 t$ about y-axis.	3 Marks 2 Marks
Q-10	Express $\int_0^1 x^m (1-x^p)^n dx$ in terms of Beta functions and hence evaluate $\int_0^1 x^{3/2} (1-\sqrt{x})^{1/2} dx$	5 Marks

Roll No.

1 2 4 1 2 0 0 3



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY SONEPAT

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Department of Information Technology (IT)

Fundamentals of Electrical and Electronics Engineering (BEL103)

End Semester Examination (Main & Re-appear)

(December 12, 2024)

Branch: IT

Semester- I

Total marks: 50

Note: All questions are compulsory

(Total pages: 3)

1. (a) Explain supermesh concept and current division rule with examples. (3)
- (b) Find the current i and the power absorbed by each element in the network as shown in **Figure- 1**. (2)
2. Define Norton's theorem. What are the steps for solving the following networks utilizing Norton's theorem. (5)
 - i) Network having only independent sources.
 - ii) Network having dependent as well as independent sources.
 - iii) Network having only dependent sources.
3. (a) Define Maximum Power Transfer (MPT) theorem. In the circuit as shown in **Figure- 2**, the maximum power transfer condition is met for the load R_L . (3)
 - a) What is the value of load resistance R_L in Ω ?
 - b) Find the maximum power through load resistance R_L in Watts.
- (b) Explain Delta-to-Star conversion in details. (2)

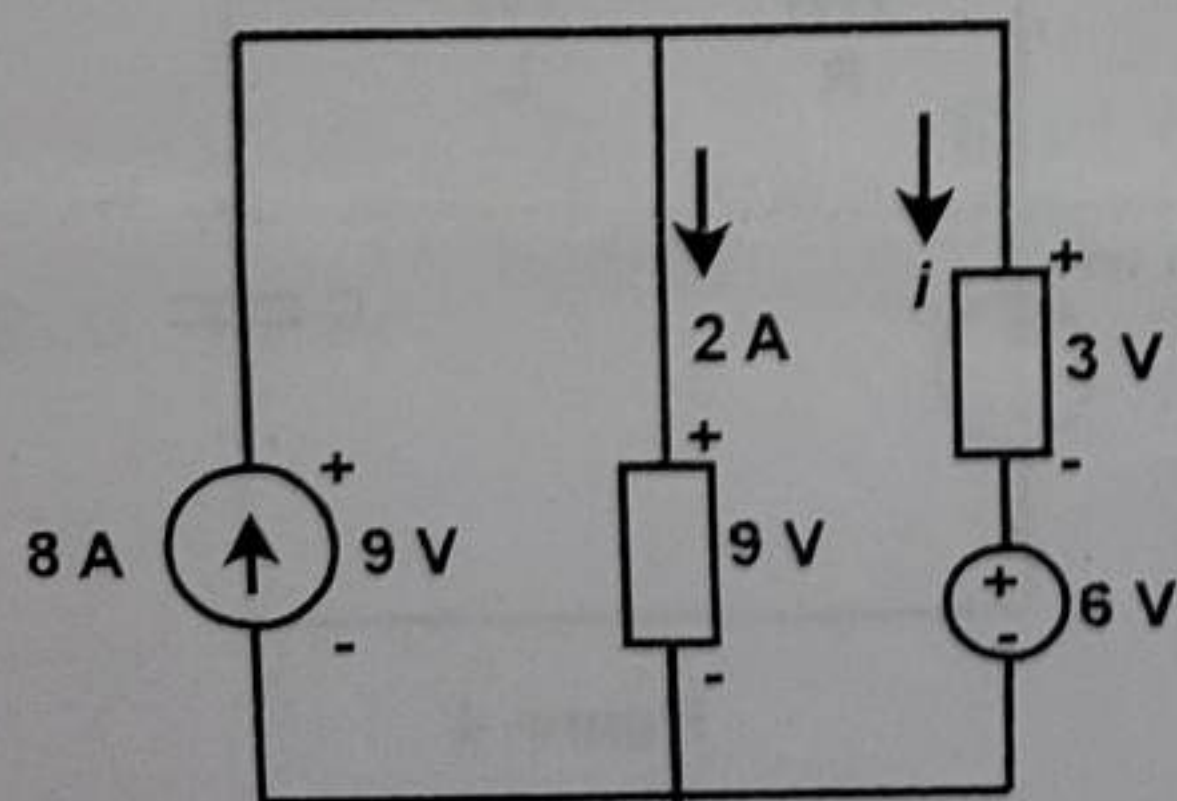


Figure- 1

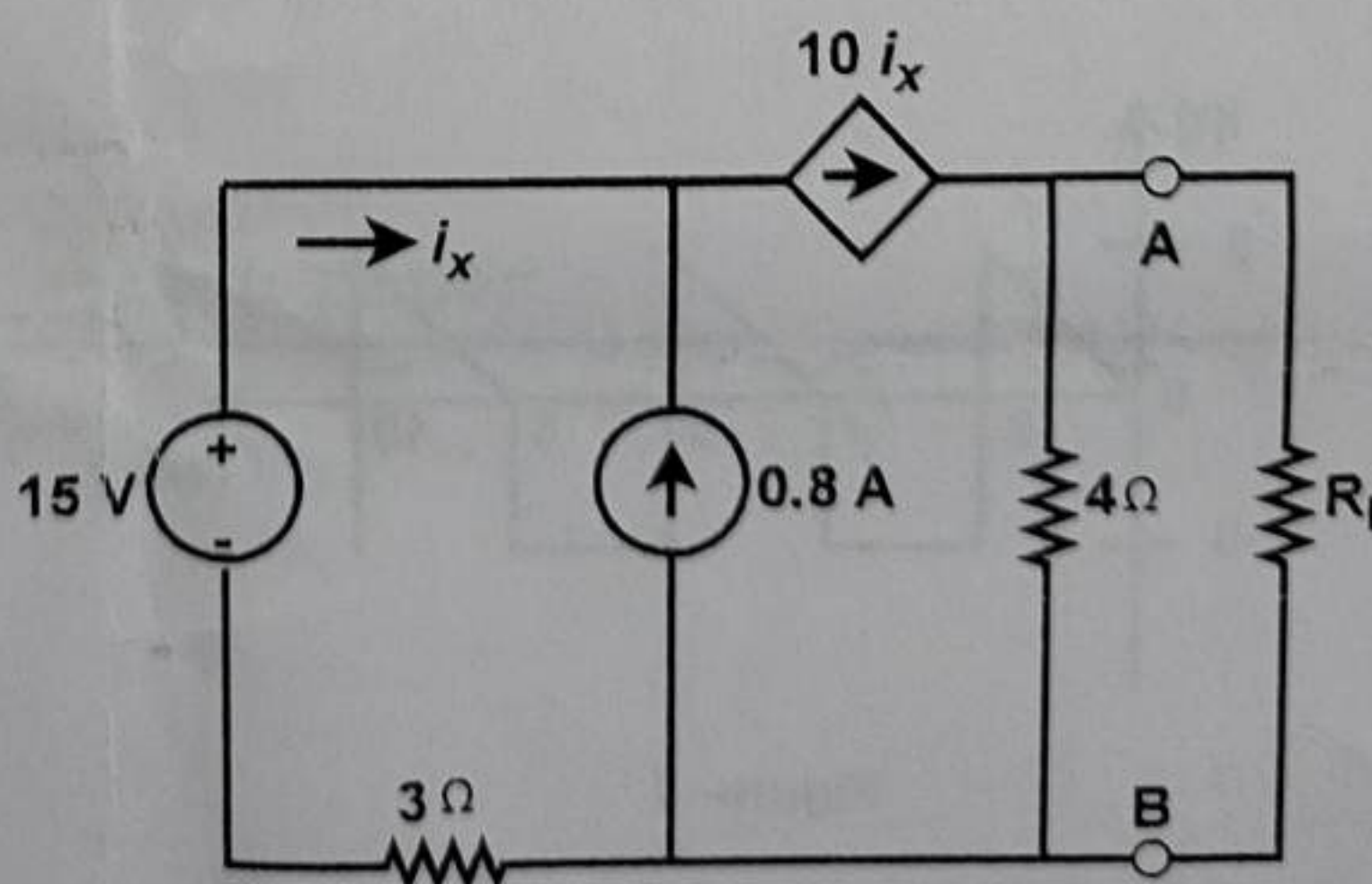


Figure- 2

4. (a) Explain a parallel RLC circuit with current and admittance phasor diagrams in details. (3)
- (b) Determine the root-mean-square (rms) value of the current waveform as shown in **Figure- 3**. If the current is passed through a $5\ \Omega$ resistor, then find the average power absorbed by the resistor. (2)
5. (a) Prove that the Quality factor $Q = \omega_o RC = \frac{R}{\omega_o L}$ is for a parallel RLC circuit. (3)
- (b) In the circuit as shown in **Figure- 4**, $R = 4\ \Omega$, $L = 2\ \text{mH}$, and $C = 0.8\ \mu\text{F}$. (2)
- i) Find the resonant frequency and the half-power frequencies.
- ii) Calculate the quality factor and bandwidth.
6. (a) A voltage source $240\angle 10^\circ\ \text{V}$ (rms), two loads $Z_1 = 30\angle -30^\circ\ \Omega$ and $Z_2 = 80\angle 45^\circ\ \Omega$ are connected in parallel. (3)
- a) Calculate the total apparent power, real power, and reactive power.
- b) What is the power factor (pf) supplied by source and seen by the source.
- (b) A voltage of $230\angle 45^\circ\ \text{V}$ (rms), 50 Hz is applied across an RLC series circuit where the input current is $10\angle -30^\circ\ \text{A}$. If $R = 5\ \Omega$, $X_L = j8\ \Omega$. Find the value of capacitive reactance in Ω and the capacitance in μF . (2)
7. Explain the following terms in details. (5)
1. Magnetic field strength and Magnetic flux density
 2. Hysteresis loss and Eddy current loss in magnetic materials
 3. Faraday's law of Electromagnetic induction
 4. Compare a magnetic circuit with an equivalent electric circuit.
 5. MMF and Reluctance
8. (a) Define Transformer and its applications. Explain the working of an ideal step-up transformer in details and prove that the ratio of induced EMF in the windings is equal to the ratio of number of turns in windings of transformer. (3)

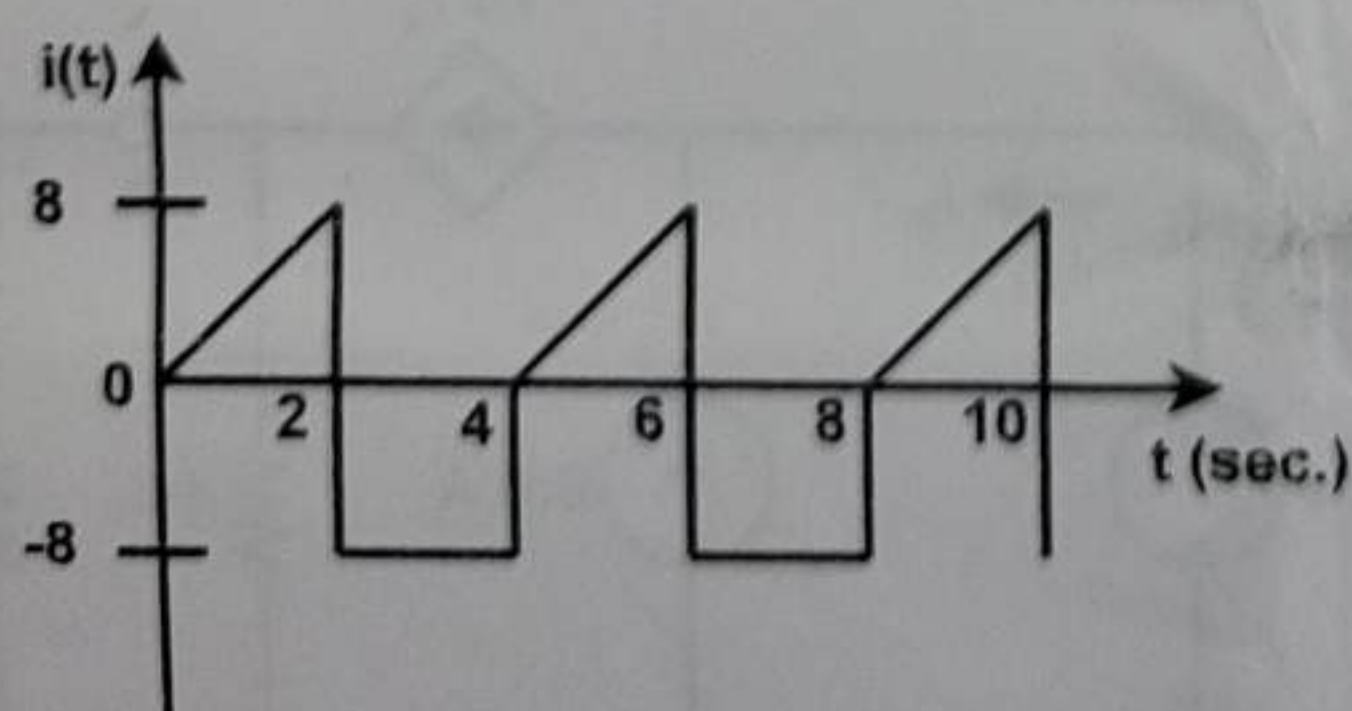


Figure- 3

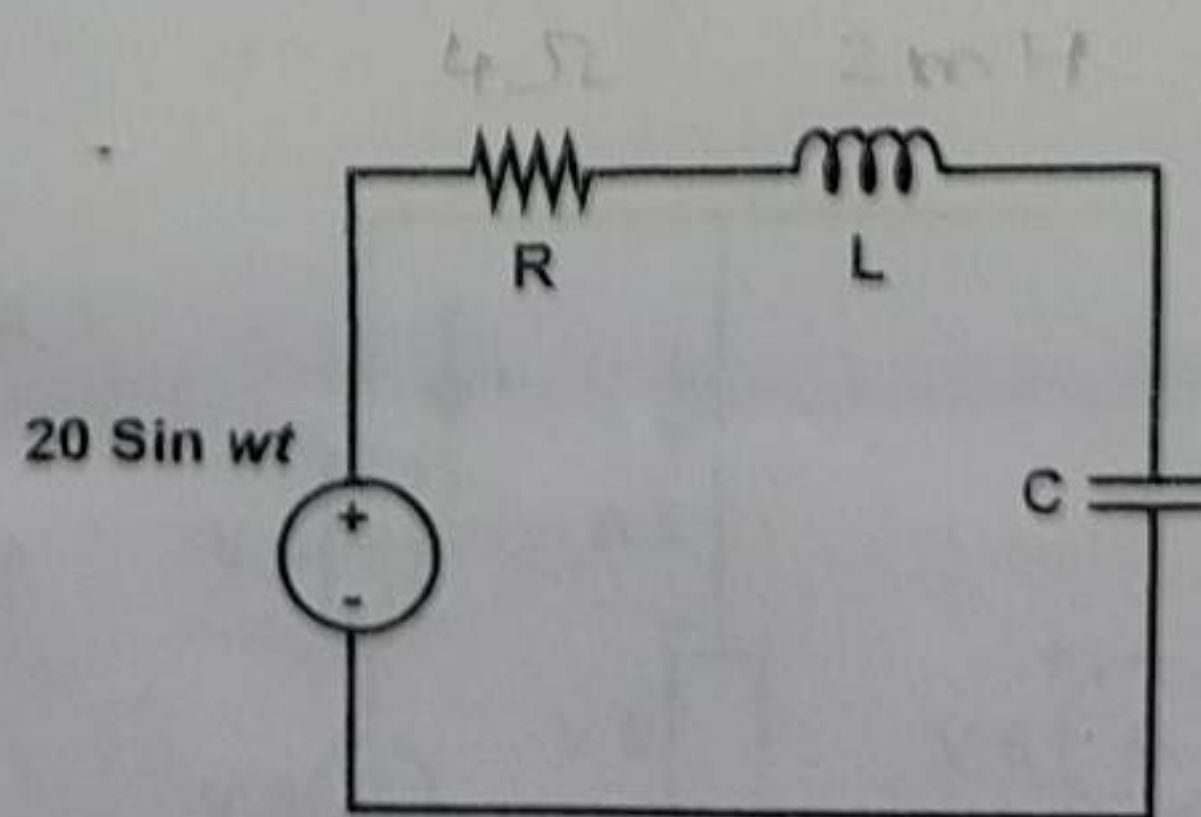


Figure- 4

- (b) A 125 V/250 V transformer is supplied with 125 V, 50 Hz supply to its low-voltage side. It is desired to have maximum value of core flux as 5.5 mWbs. Calculate the required number of turns in its primary winding. (2)

9. (a) Define the following terms of P-N junction diode. (3)

- Drift and Diffusion Current
- Depletion region and Reverse breakdown voltage
- A silicon diode is connected across a 5 V supply with a series resistance of $40\ \Omega$ as shown in **Figure- 5**. Neglecting diode resistance, calculate the diode current.

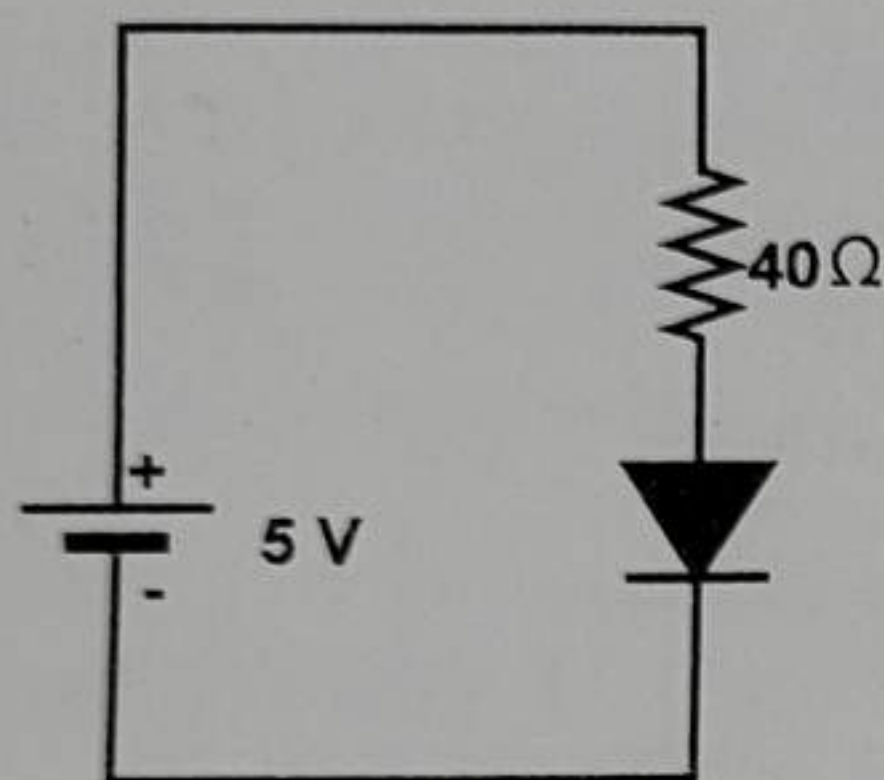


Figure- 5

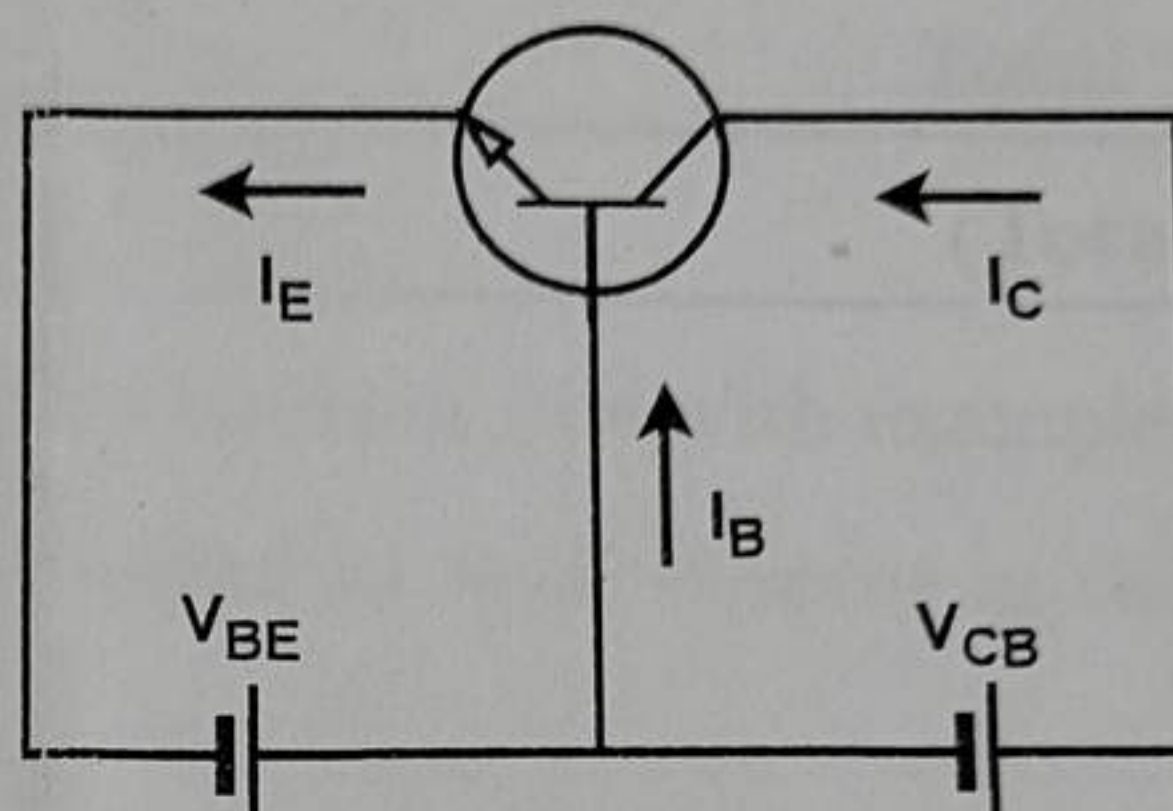


Figure- 6

- (b) Explain the working of forward-biased P-N junction diode in details. Also draw the V-I characteristics of diode. (2)

10. (a) Define a Transistor and its applications. Explain the working of n-p-n transistor for common base configuration. Also draw its input and output characteristics. (3)

- (b) An n-p-n transistor as shown in **Figure- 6** has been provided with biasing voltage V_{BE} and V_{CB} . Calculate the values of I_C and I_E if α is 0.98 and I_B is $60\ \mu A$. Also find the value of β_{dc} . (2)

End of question paper



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End Sem Exam

Subject: Analog Electronics (ECL-106)

Roll no. 12412003

Branch: IT

M.M.-50 marks

Time- 3 Hrs.

Note: All questions are compulsory.

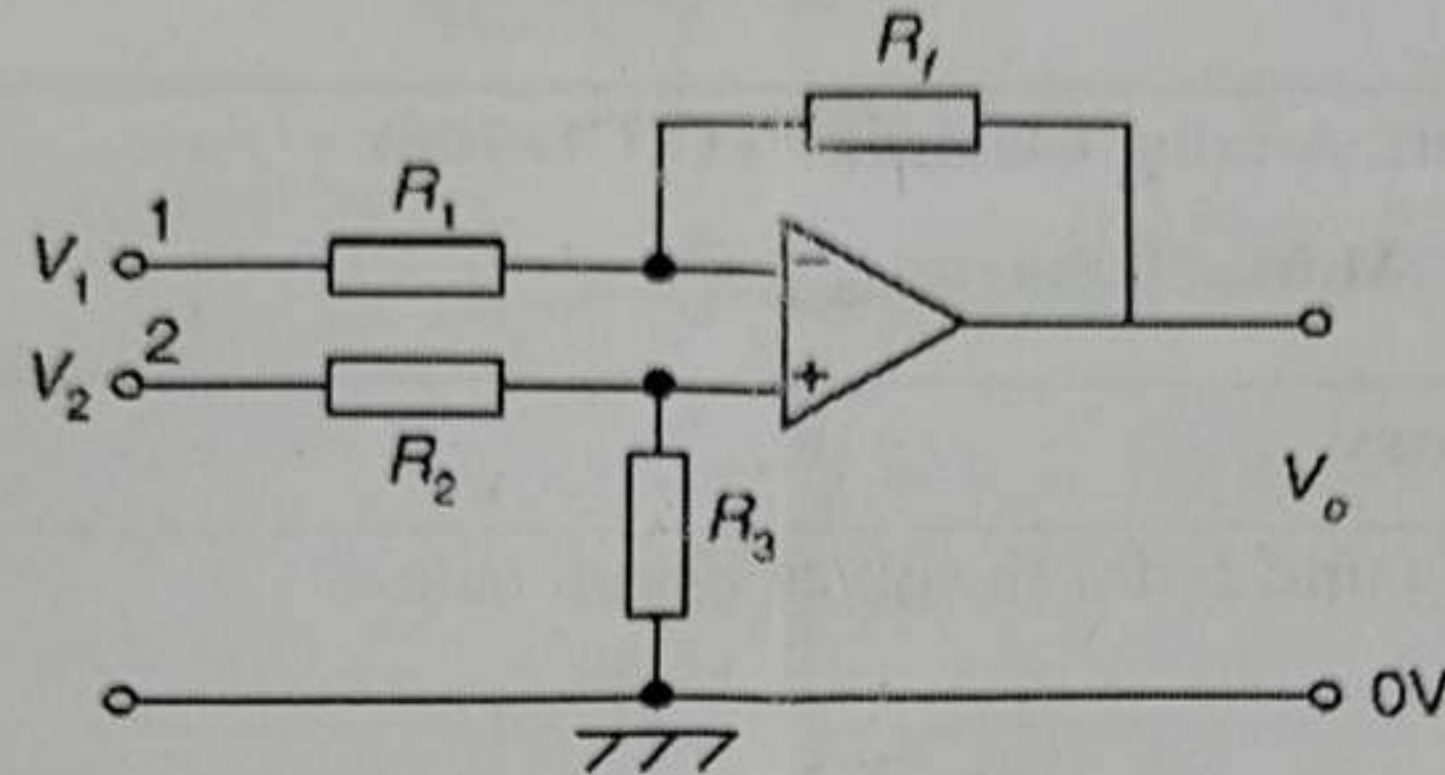
Q-1:(a)	Determine V_D , V_R , current I_D and I_R for the given circuit below.	(5 marks)
(b)	A Zener diode with a breakdown voltage of $V_Z = 5.6$ V is used in a voltage regulator circuit. The input voltage is $V_{in} = 20$ V, and the series resistor $R_S = 330\Omega$. Calculate the Zener current I_Z when the load current $I_L = 20$ mA.	(5 marks)
Q-2:(a)	Differentiate between NPN and PNP transistors in terms of structure and operation in active model.	(5 marks)
(b)	For a CE amplifier circuit, calculate the values of resistors R_B and R_C when $I_C = 2$ mA and $V_{CE} = 5$ V, given $V_{CC} = 12$ V, $\beta = 100$ and $V_{BE} = 0.7$ V.	(5 marks)
Q-3:(a)	Illustrate the symbols for N-channel and P-channel enhancement-type MOSFETs, clearly labeling all terminals. A MOSFET has a threshold voltage $V_{th} = 2$ V. If $V_{GS} = 5$ V and $V_{DS} = 10$ V, determine whether the MOSFET is in saturation or linear region.	(5 marks)
(b)	Sketch the transfer and drain characteristics of an n-channel depletion-type MOSFET with $I_{DSS} = 12$ mA and $V_P = -4$ V for a range of $V_{GS} = V_P$ to $V_{GS} = 1$ V.	(5 marks)
Q-4:(a)	Explain Colpitts oscillator and derive the equation for oscillation.	(5 marks)
(b)	In a Wien Bridge oscillator, the input resistance of amplifier is $R_1 = 5$ kΩ and feedback resistance of amplifier is $R_2 = 10$ kΩ. Verify if the gain condition for sustained oscillations is satisfied with proper explanation.	(5 marks)
Q-5:(a)	Design a summing amplifier using an op-amp that sums three input voltages $V_1 = 1$ V,	(5 marks)

$V_2 = 2\text{ V}$, and $V_3 = 3\text{ V}$, and produces an output voltage $V_o = -10\text{ V}$. Find the feedback resistor value, if all input resistors R_1 , R_2 , R_3 are equal to $5\text{ k}\Omega$.

(b)

In the differential amplifier below, $R_1 = R_2 = 10\text{ k}\Omega$, and $R_3 = R_f = 100\text{ k}\Omega$. Determine the output voltage V_o if $V_1 = 25\text{ mV}$ and $V_2 = 50\text{ mV}$.

(5 marks)



*****Best Wishes*****



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End sem-1

Subject: Environmental Studies

Sub code:(HUL 102)

Roll no.....0.3....

Branch: IT

M.M.-50 marks

Time- 3 hr.

Note: All questions are compulsory.

Q-1(A)	What is a natural resource and differentiates between renewable and non-renewable resources?	(5 marks)
(B)	Explain the significance of forests as a natural resource.	(3 marks)
Q-2	How is primary ecological succession different from secondary ecological succession?	(2 marks)
Q-3(A)	What is biodiversity and how can we conserve it?	(3 marks)
(B)	Define genetic and species biodiversity.	(2 marks)
Q-4	Explain in-situ and ex-situ conservation of biodiversity	(5 marks)
Q-5(A)	Define pollution, what is National Air quality index.	(3 marks)
(B)	What are the causes, effects and control measures of Air pollution?	(2 marks)
Q-6(A)	What are the causes, effects and control measures of industrial wastes.	(3 marks)
(B)	Differentiate solid waste depending upon their generation.	(2 marks)
Q-7(A)	What is greenhouse effects, explain global warming potential of greenhouse gases?	(3 marks)
(B)	What is sustainable development.	(2 marks)
Q-8	What is rainwater harvesting? Explain construction of rain water harvesting system.	(5 marks)
Q-9(A)	Explain the role of value education on population and environment.	(3 marks)
(B)	Explain the consequence of uncontrolled population growth on environment.	(2 marks)
Q-10	Explain the role of information technology in environment and human health	(5 marks)



IT & DSA END TERM EXAMINATION (December 2024)

Course Name: Health, Sports & Safety

Course Code: SAP-102

Total Marks: 50

Total Duration: 3 Hour

QUESTION PAPER

Each question Carries 5 marks.

Question 1: What is Explosive strength? Explain the Push-up Test?

Question 2: What is Ashtanga Yoga? Explain Yama and its components?

Question 3: Explain Fartlek Training method?

Question 4: What is scoliosis, its causes and remedies?

Question 5: How to perform Top to Toe assessment of the casualty?

Question 6: What is CPR? Explain How to perform CPR?

Question 7: What is BMI and how it is calculated? Calculate your score?

Question 8: What are the steps of dealing with an emergency situation to provide First aid?

Question 9: What is Coordinative ability? Explain any of its 2 components?

Question 10: What is First Aid and its aims?



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Branch: IT

Subject Name: Computer Programming (CP)

Subject code: UIT105

Semester 1st

Mid Sem-EST

M.M.-50 marks

Time- 3 Hrs.

Roll no

Note: All questions are compulsory. Solve all question and the part of the question in order.

Q.1 A Draw a comparative analysis table to explain all the storage classes for their storage location, life and scope. (3 marks)

B. A C program contains the following declarations and initial assignments (5 marks)
int i=8, j=5;
float x=0.005, y= -0.01 ;
char c='c', d='d';

Identify the value of each of the following

1. printf("%d\n", (3*i-2*j)%(2*d-c));
2. printf("%d\n", (i-2*j)%(c+2*d)/(x-y));
3. printf("%d\n", 2*((i/5)+(4*(j-3))%(i+j-2)));
4. printf("%d\n", (x>y)&&(i>0)&&(j<6));
5. printf("%d\n", (x>y)&&(i>0)||j<6));

Q.2 A What are the most important features of C Language? (3 marks)

B. Write a program to check whether a person is eligible to vote based on their age. (5 marks)

Q.3 A. Write a C Program to print the following pattern using for loop: (3 marks)

1
11
111

B. Why do we need dynamic memory allocation? Explain the functioning of calloc, realloc and free with suitable code. (5 marks)

Q.4 A. What are the differences between the following: (3 marks)

1. ++i and i++
2. Recursion and Iteration
3. structure and union

B. Write a program to declare time structure and read two different time period and display sum of it. (5 marks)

Q.5 A. Write a c program to find the addition of two numbers using macros. (3 marks)

B. Why is recursion not always the optimal choice for solving problems, despite its ability to simplify code? Justify your answer with examples of scenarios where recursion leads to poor performance or inefficiency. (5 marks)

Q.6 Write short note on the followings: (10 Marks)

1. Bitwise Operator
2. strcpy()
3. Return -1
4. User defined datatype
5. Nested Loops

End



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End Sem. Exam

Subject : Applied Science

Code: BSL104

Roll no

Branch: IT

Year: 2024

M.M. 50 marks

Time-3 hrs

Note:

- Attempt all 5 questions in section A and any 2 questions in Section B
- Calculator is allowed

Section A (5*6 marks)

1. The angular frequency for a wave propagating inside a waveguide is given in terms of the wave number k and the width b of the waveguide by

6 marks

$$\omega = kc \left[1 - \frac{\pi^2}{(b^2 k^2)} \right]^{-\frac{1}{2}}. \text{ Find the phase and group velocities of the wave.}$$

2. Show that the wavelength of a matter-wave associated with a particle of rest mass m_0 and kinetic energy K , is given by

6 marks

$$\lambda = \frac{hc}{\sqrt{K(K + 2m_0 c^2)}}$$

3. Suppose that the wave function of a quantum mechanical particle in an infinite potential box of length L is given by $\psi(x) = A x (L - x)$ for $0 \leq x \leq L$, and $\psi(x) = 0$ elsewhere. Find the normalization constant A , the expectation values of momentum, $\hat{p}$ and Hamiltonian operator, $\hat{H}$

6 marks

4. A copper crystal has an fcc lattice with one Cu atom per lattice point and a lattice constant of $3.61 \text{ \AA}$. Every copper atom has one valence electron. Estimate the electron Fermi energy, Fermi velocity and electron wavelength in a copper crystal.

6 marks

5. The edge length of face centered cubic unit cell of a certain material is 'a'. The planar packing fraction (ppf) is defined as the ratio of area occupied

6 marks

by effective number of atoms to the area of a plane. Draw a plane having Miller indices given as (110) plane and deduce an expression of ppf for this plane.

Section B (2*10 marks)

6. Describe the seven crystal systems of Bravais lattice with suitable diagrams. **10 marks**
7. Derive the expression of Fermi energies of n and p type semiconductors and comment on their dependence on temperature and impurity concentrations. **10 marks**
8. Discuss all the physical and chemical properties of nano materials with proper reasoning. **10 marks**